

Government Engineering College Kishanganj

Department of Computer Science and Engineering
Computer Networks LAB-02 — Exploring Packet Sniffer and Packet Analyzer
Answer Sheet | Capture file analyzed: labb.pcapng

Q1. [3 marks] Protocols observed at each layer

Layer	Protocols Observed
A. Application layer	DNS (name resolution), HTTP (neverssl.com — cleartext), TLS/HTTPS (cornell.edu — TLS 1.3), OCSP (automatic certificate-revocation check triggered by the TLS handshake)
B. Transport layer	TCP (both HTTP and HTTPS connections), UDP (carries DNS queries)
C. Network layer	IPv4 and IPv6 (both used — both sites have A and AAAA records), ARP (local address resolution)

Note: Background noise (QUIC, mDNS, SSDP, IGMP, ICMPv6) was also observed in the capture, originating from Windows/system background processes running during capture — not part of the neverssl.com / cornell.edu exercise traffic itself.

Q2. [2 marks] Total data received

Site	Total Data Received	Evidence
A. http://neverssl.com/	4,261 bytes (HTTP response: headers + body)	Server sent data across 8 TCP segments, Seq 1 → 4261 (server FIN packet has Seq=4262, confirming 4261 bytes sent)
B. https://www.cornell.edu	89,212 bytes (TLS-encrypted application data)	Server FIN packet has Seq=89213, confirming bytes 1–89212 were sent by the server

Cornell.edu's data is TLS-encrypted, so only the total encrypted byte count is measurable from the capture — the actual HTML content is not visible. This illustrates the core visibility difference between HTTP and HTTPS traffic.

Q3. [6 marks] Deep-dive on <http://neverssl.com/>

A. HTTP GET requests observed

Exactly **one** HTTP GET request was observed (curl fetches only the base URL, not any embedded resources):

```
GET / HTTP/1.1
Host: neverssl.com
User-Agent: curl/8.21.0
Accept: */*
```

B. HTTP server version

From the response header: **Server: Apache/2.4.66 ()**, protocol **HTTP/1.1**, status **200 OK**.

C. TCP segments and data volume for the response

Metric	Value
(i) Total TCP segments received	8 segments (frames 3610, 3611, 3612, 3615, 3617, 3619, 3620, 3621)

(ii) Total data in HTTP response	4,261 bytes total (headers ≈ 300 bytes + body — Content-Length: 3961 bytes)
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D. TCP Three-Way Handshake

Client (port 16234) ↔ Server 34.223.124.45:80

Step	Frame	Direction	Flags	Seq (relative)	Seq (absolute / raw)
SYN	3126	Client → Server	SYN	Seq=0	Seq=3299535659
SYN-ACK	3589	Server → Client	SYN, ACK	Seq=0, Ack=1	Seq=2535418276, Ack=3299535660
ACK	3591	Client → Server	ACK	Seq=1, Ack=1	Seq=3299535660, Ack=2535418277

Note: The client's SYN was retransmitted 4 times (frames 3126, 3134, 3236, 3569) before the server responded — the server was slow to respond initially. Both relative (Wireshark's default, zero-based) and absolute (raw) sequence numbers are provided above.